

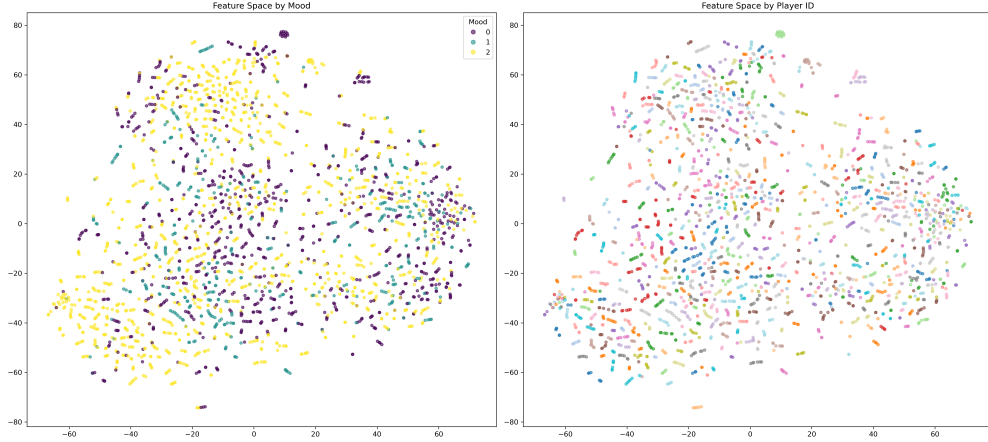


Figure 1: t-SNE visualization for Solid (Fold 1). Points are colored by Player ID, showing high intermixing across players. Similar patterns were observed in all 9 games and across all 5 folds.

Please note that additional detailed Supplementary Materials, owing to their substantial file size, have been hosted in a separate, dedicated Supplementary Materials repository to facilitate access: <https://tinyurl.com/254je2ct>.

1 data split evidence examples

Here, we provide supplementary evidence supporting the data split protocol described in Section 4.2 of the main paper. All analyses are based on the Solid dataset as an example shown here. The evidence of other game datasets can be found in our supplementary repository.

1.1 t-SNE Visualization of Feature Embeddings

Figure 1 shows t-SNE visualizations of feature embeddings extracted from the Solid test set using the full fine-tuned ViViT model, colored by Player ID. The consistent intermixing across players in all games supports our claim that player identity does not create distinct clusters in the feature space.

1.2 Distribution of Intra-User Label Entropy

Figure 2 presents the distribution of intra-user label entropy across all players in Solid. The mean entropy of 0.75 confirms high label diversity within individual players’ sessions.

2 Empirical Validation of the Selection Criterion

To assess the reliability of our proposed L2 selection criterion, we conducted a preliminary study comparing it against the exhaustive cross-validation (CV) oracle used in Surgical Fine-Tuning (SFT) [1]. This SFT-CV protocol, which involves individually fine-tuning each candidate layer (or block) to select the best-performing one based on validation performance, was shown in the original work to be the most effective selection strategy, outperforming their proposed heuristic criteria (RGN/SNR).

Protocol. We implemented this SFT-CV protocol on a subset of games under a simpler data split for computational feasibility. Each candidate layer (alongside the classifier) was fine-tuned for a short budget of 3 epochs—a duration we found sufficient to observe significant performance differentiation. The layer achieving the highest weighted F_1 -score on a held-out validation set was selected as the CV oracle choice.

Result. As shown in Table 1 and Table 2, the single layer selected by our L2 criterion during SALFT’s exploratory phase **exactly matched the oracle layer identified by the exhaustive SFT-CV protocol**

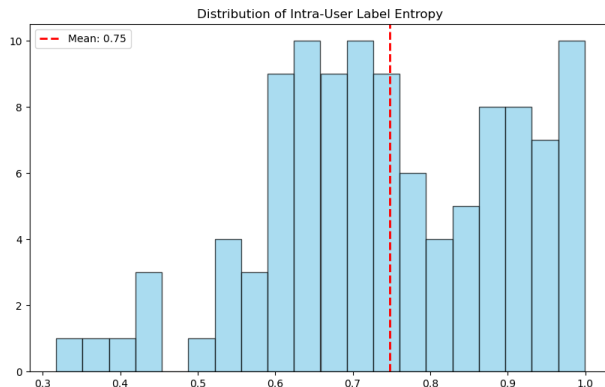


Figure 2: Distribution of intra-user label entropy. The mean of 0.75 indicates substantial label diversity within individual players’ sessions.

Table 1: Cross-val results of game performance by layer (weighted F_1). All values are percentages. Best results per game are in **bold**.

Layer	apex	endless	fps	gallery	gun	platform	solid	tiny	topdown
embeddings	50.70	46.23	58.15	45.92	49.10	41.29	58.58	48.39	53.40
layer.0	72.97	61.49	72.67	65.41	68.24	72.88	73.19	64.97	71.46
layer.1	65.79	55.84	71.81	62.07	56.28	60.18	72.86	55.79	68.42
layer.2	56.06	53.30	68.08	61.97	51.90	62.58	70.01	54.41	62.59
layer.3	56.09	52.36	69.70	58.68	50.09	54.73	65.30	51.49	61.17
layer.4	56.28	50.67	68.18	58.17	48.49	52.16	62.25	50.49	61.35
layer.5	54.02	53.48	67.66	59.50	50.03	52.88	60.86	54.21	61.87
layer.6	48.30	53.68	70.16	55.80	48.79	54.68	61.38	54.03	61.47
layer.7	49.30	54.11	70.20	60.79	50.08	52.02	62.13	53.78	63.69
layer.8	47.76	51.70	70.23	56.55	52.07	55.56	62.40	52.80	62.75
layer.9	52.27	53.84	72.36	61.06	52.90	54.57	64.08	53.41	63.54
layer.10	50.00	54.72	70.56	58.52	51.76	58.73	67.22	54.25	64.96
layer.11	47.96	50.57	67.15	52.89	48.31	49.25	57.92	52.86	60.46
classifier only	40.27	41.12	55.18	40.79	37.94	42.13	46.71	45.43	50.26

in all tested cases. This perfect alignment provides strong empirical evidence that our efficient, single-run L2 metric is a reliable proxy for the computationally expensive optimal selection procedure.

3 Statistical Significance Tests of SALFT

To rigorously evaluate our method, we conduct exact two-sided Wilcoxon signed-rank tests for each game and evaluation metric, comparing SALFT to: (1) fully fine-tuned ViViT and (2) the best-performing baseline per game and metric. Given the limited sample size of 5-fold cross-validation ($N = 5$), the mathematically minimum possible p -value is 0.0625, precluding the use of traditional significance thresholds (e.g., $\alpha = 0.05$). Therefore, we report p -values and mean differences (Diff), treating instances that achieve this theoretical minimum ($p = 0.0625$) as evidence of consistent empirical superiority across all five folds. A positive Diff indicates that SALFT outperforms the compared method.

4 Complete Computational Efficiency Analysis

In this section, Figure 3 shows the complete set of computational efficiency curves for rest eight games, complementing the representative example shown in Figure 2 of the main paper. These curves consistently

Table 2: Cross-validation results of game performance by layer (macro F_1). All values are percentages. Best results per game are in **bold**.

Layer	apex	endless	fps	gallery	gun	platform	solid	tiny	topdown
embeddings	44.0	40.76	52.23	32.8	42.61	32.13	50.68	38.43	44.67
layer.0	71.18	58.8	69.91	60.75	65.29	71.24	69.54	62.71	67.61
layer.1	63.83	52.01	68.22	55.72	50.61	54.87	67.29	48.38	65.29
layer.2	50.7	49.72	63.57	55.85	44.74	58.62	66.62	46.5	57.36
layer.3	50.84	48.07	65.93	51.41	42.95	50.07	59.36	44.36	54.98
layer.4	51.55	45.96	63.96	50.15	40.23	45.03	55.31	43.04	54.86
layer.5	48.92	49.35	63.34	52.65	42.5	47.81	54.27	46.67	56.16
layer.6	42.0	50.01	66.62	47.16	40.26	50.07	54.9	46.67	55.55
layer.7	41.95	50.7	66.42	54.21	42.68	45.58	56.92	46.92	58.54
layer.8	40.97	47.56	66.3	48.25	45.12	49.95	56.91	44.63	57.67
layer.9	46.63	50.52	68.99	54.34	45.87	49.42	59.62	45.57	58.91
layer.10	43.99	51.08	66.8	50.99	44.86	53.98	62.32	46.81	60.43
layer.11	40.22	46.33	62.96	43.42	40.65	42.72	51.25	45.34	54.91
classifier only	30.93	34.06	45.63	24.07	27.24	32.87	34.53	35.46	39.65

Table 3: Detailed statistical results of the exact two-sided Wilcoxon signed-rank test comparing SALFT with fully fine-tuned ViViT and the best-performing baseline, grouped by game. Due to the sample size of 5-fold cross-validation ($N = 5$), traditional significance levels (e.g., $p < 0.05$) cannot be reached. Instead, the theoretical minimum possible p -value is 0.0625, which indicates empirical superiority across all 5 folds. Instances reaching this theoretical minimum are highlighted in **bold**.

Game	Metric	Ours (Mean $\pm$ Std)	vs Full		vs Base	
			p -value	Diff	p -value	Diff
Apex	fl_weighted	0.5845 $\pm$ 0.0070	0.8125	0.0040	0.8125	0.0040
	fl_macro	0.5400 $\pm$ 0.0122	0.6250	-0.0044	0.6250	-0.0044
	accuracy	0.5913 $\pm$ 0.0071	0.6250	0.0093	0.6250	0.0093
Endless	fl_weighted	0.5558 $\pm$ 0.0137	0.0625	0.0104	0.1875	-0.0269
	fl_macro	0.5149 $\pm$ 0.0179	0.4375	0.0091	0.1250	-0.0340
	accuracy	0.5634 $\pm$ 0.0147	0.1250	0.0141	0.1875	-0.0213
FPS	fl_weighted	0.6275 $\pm$ 0.0139	0.8125	0.0038	0.6250	0.0038
	fl_macro	0.5799 $\pm$ 0.0158	0.8125	0.0053	0.8125	0.0026
	accuracy	0.6358 $\pm$ 0.0133	0.6250	0.0055	0.6250	0.0043
Gallery	fl_weighted	0.5669 $\pm$ 0.0162	0.1250	0.0141	0.1250	0.0141
	fl_macro	0.4958 $\pm$ 0.0197	0.0625	0.0234	0.0625	0.0234
	accuracy	0.5858 $\pm$ 0.0170	0.6250	0.0072	0.6250	0.0072
Gun	fl_weighted	0.5727 $\pm$ 0.0130	0.0625	0.0312	0.0625	0.0312
	fl_macro	0.5408 $\pm$ 0.0131	0.0625	0.0324	0.0625	0.0324
	accuracy	0.5778 $\pm$ 0.0111	0.0625	0.0283	0.0625	0.0283
Platform	fl_weighted	0.5639 $\pm$ 0.0243	0.1250	0.0202	0.1250	0.0202
	fl_macro	0.5308 $\pm$ 0.0253	0.1875	0.0218	0.1875	0.0218
	accuracy	0.5702 $\pm$ 0.0232	0.0625	0.0204	0.0625	0.0204
Solid	fl_weighted	0.6058 $\pm$ 0.0147	0.3125	0.0115	0.3125	0.0115
	fl_macro	0.5453 $\pm$ 0.0146	0.6250	0.0023	0.6250	0.0023
	accuracy	0.6141 $\pm$ 0.0137	0.1875	0.0161	0.1875	0.0161
Tiny	fl_weighted	0.5560 $\pm$ 0.0187	0.6250	-0.0075	0.6250	-0.0075
	fl_macro	0.5211 $\pm$ 0.0256	0.6250	-0.0128	0.6250	-0.0128
	accuracy	0.5647 $\pm$ 0.0158	0.6250	-0.0034	0.6250	-0.0034
Topdown	fl_weighted	0.5974 $\pm$ 0.0120	0.6250	0.0083	0.4375	-0.0036
	fl_macro	0.5495 $\pm$ 0.0135	0.8125	0.0042	0.1875	-0.0113
	accuracy	0.6038 $\pm$ 0.0109	0.6250	0.0134	1.0000	0.0025

demonstrate that SALFT achieves performance comparable to full fine-tuning with significantly lower cumulative computational cost across all games.

5 Comparison of Attention Patterns: SALFT vs. Full Fine-Tuning

To further understand the behavioral differences between our proposed SALFT method and standard full fine-tuning, we compare their attention patterns using representative samples from the `solid` game. Both models analyzed in this section are from the first fold of the 5-fold cross-validation.

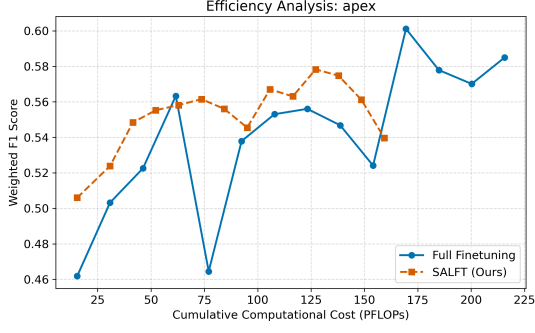
As shown in Figure 4, the model fine-tuned via full fine-tuning exhibits a strong tendency to focus **narrowly** on specific, well-defined regions (e.g., a leading vehicle or lane boundaries), producing sharply localized saliency maps. While this indicates precise attention to salient objects, it may limit the model’s capacity to integrate broader contextual cues from the scene.

In contrast, the SALFT-fine-tuned model (see the Figure 3 of main paper) demonstrates a slightly broader attentional spread. It covers not only primary objects of interest but also incorporates relevant surrounding context, such as road edges, background structures, and indicators of camera motion. This pattern appears better suited for interpreting the complex, dynamic scenes characteristic of gameplay videos.

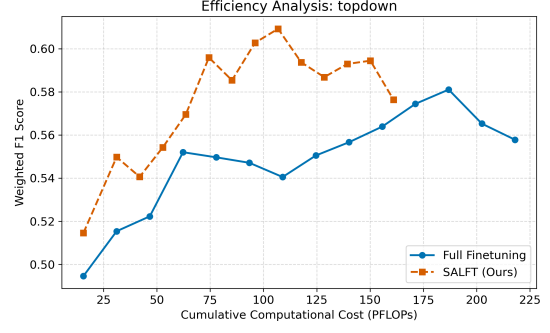
This divergence in attention strategy may underlie the performance differences noted in the main paper. The broader, context-integrating attention of the SALFT model likely contributes to its robust performance across most games. Furthermore, we observed that the full fine-tuned model occasionally showed reduced sensitivity to subtle emotional transitions, misclassifying some arousal-decreasing segments as arousal-unchanging, whereas the SALFT model maintained better discriminability in these cases. Additional attention maps for other games and arousal classes, along with further comparative analyses, are available in our online repository.

References

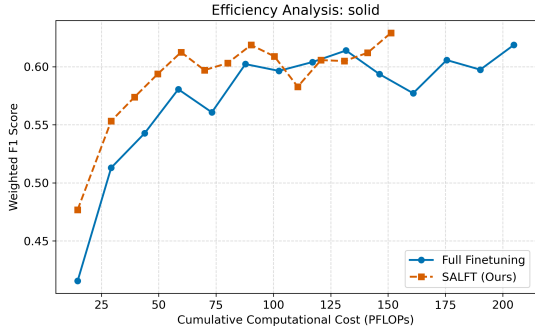
- [1] Y. Lee et al., “Surgical fine-tuning improves adaptation to distribution shifts,” in *The Eleventh International Conference on Learning Representations*, 2023. [Online]. Available: <https://openreview.net/forum?id=APuPRxjHvZ>.



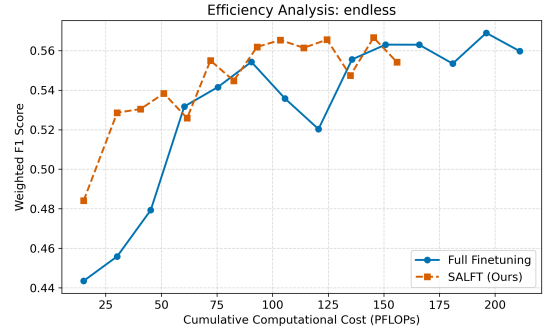
(a) apex



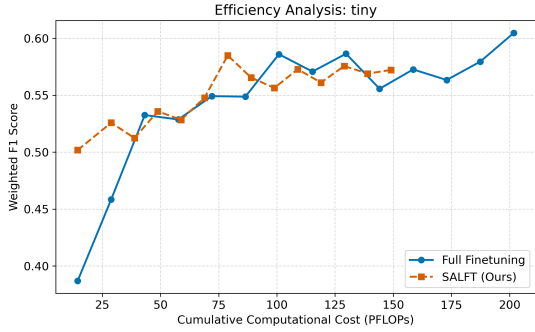
(e) topdown



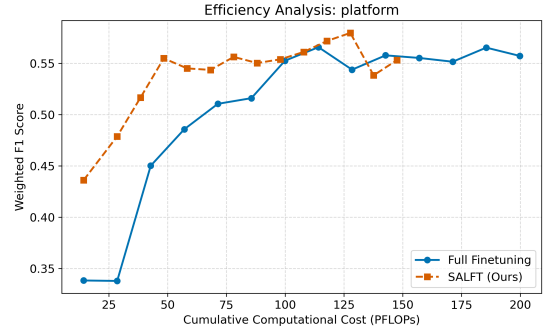
(b) solid



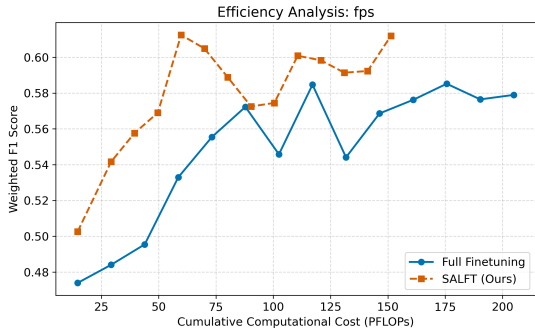
(f) endless



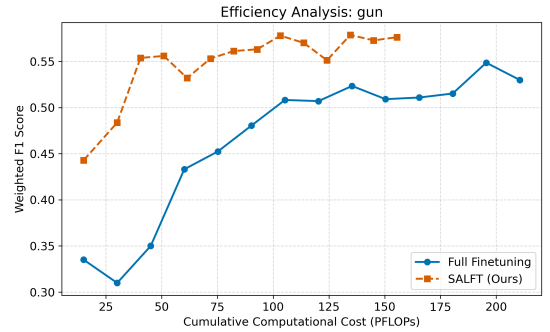
(c) tiny



(g) platform



(d) fps



(h) gun

Figure 3: Complete set of weighted F_1 against cumulative fine-tuning FLOPs for all nine games (full fine-tuning vs. SALFT). The two-stage nature of SALFT is evident in all curves, with an initial exploratory phase (first 2 epochs) followed by the efficient selective fine-tuning phase.

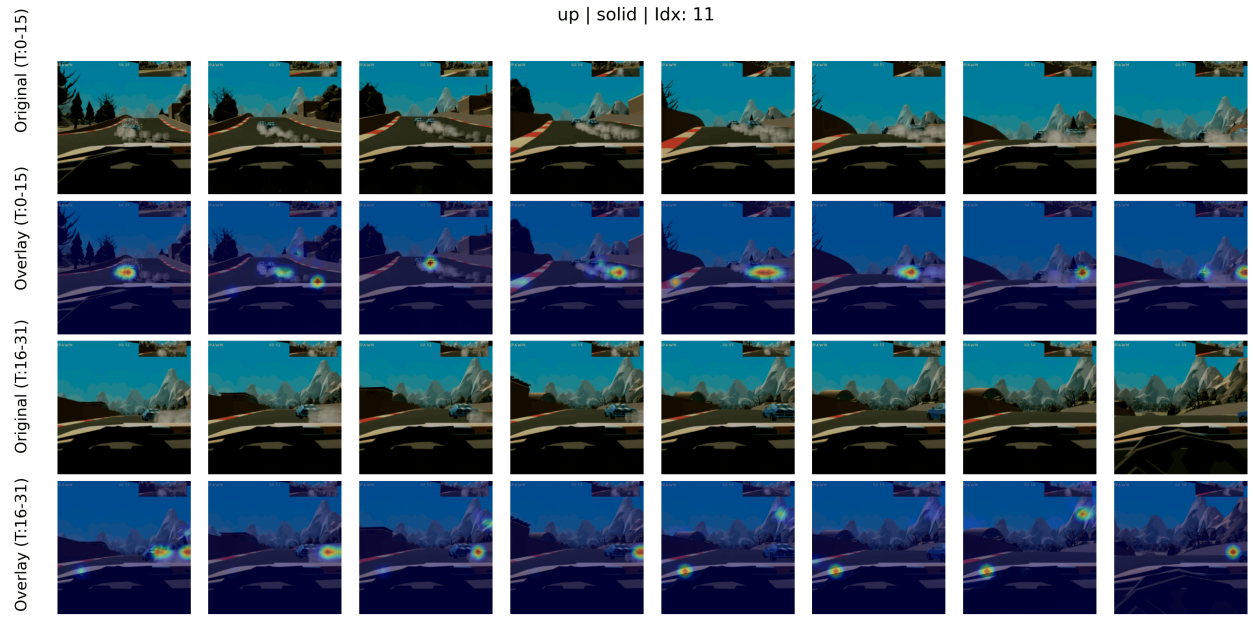


Figure 4: Interpretation example for an arousal-increasing segment in **solid**, generated by the model fine-tuned via **full fine-tuning** (first fold).